

Response Letter with Annotations

Zhou, A., Yang, T., & González-Bailón, S. (2023). *The puzzle of misinformation: Exposure to unreliable content in the United States is higher among the better informed*. *New Media & Society*, 27(3), 1526-1543. <https://doi.org/10.1177/14614448231196863>

Annotations

1. Purposes

- a. Motivation (one-sentence summary)
 - What other news do people who engage with misinformation consume, and how does misinformation fit into their broader news consumption?
- b. Specific Rationale (several bullet points)
 - Previous misinformation research focused on measuring the prevalence and reach of misinformation; demographic characteristics of misinformation's likely audiences and the interventions to combat misinformation.
 1. However, misinformation is not as widespread as public discourse often suggests, that its sharing is driven by demographic factors such as ideology and age, and that although people can generally tell higher-quality sources from lower-quality ones, labeling content as false does not substantially reduce misperceptions.
 - Emerging evidence indicates that individuals who consume unreliable content also consume substantial amounts of reliable news, suggesting that exposure to misinformation may be embedded within broader news diets rather than replacing them.
 1. However, the relationship between exposure to unreliable content and the ideological diversity and intensity of individuals' overall news diets remains underexplored.
- c. Research Questions and/or Hypotheses
 - Identify IV, DV, mediator, moderator
 1. **DV**
 - a. Exposure to unreliable content/misinformation (count of pages visited; time spent on unreliable sources)
 2. **IVs**
 - a. Overall news consumption/interest in the news.
 - b. Ideological diversity of news domains.
 3. **Controls:**
 - a. Demographic variables (age, gender, education, income, race, employment)-as fixed effects.
 - b. Partisan leaning-additional robust check/control.
 - Identify specific relationships among variables
 1. The study tests whether **interest in the news** (news pages accessed) and **ideological diversity of news diets** predict higher **exposure to unreliable content**, net of demographic covariates (and robust to partisan leaning).

2. Method

a. Research Design

- Observational, longitudinal panel study using passive web-browsing data (not an experiment or self-report survey).

b. Sample (representative sample?)

- $N \sim 140,000$ unique users in the United States for the period of January–December 2018.
- Measurement company: Nielsen
- Weighted to be representative of the US population

c. Measures

- Measurement of **exposure to unreliable content**: count of pages visited; time spent on unreliable sources.
 1. Measurement of **unreliable sources**: Unreliable sources are identified at the domain level by combining an established misinformation domain list with NewsGuard's credibility ratings. Websites scoring below 60 on NewsGuard's 0–100 scale are classified as unreliable.
- Measurement of **exposure to reliable content**: authors merged the lists of news sources used in five previous published studies to define mainstream news exposure.
- Measurement of **interest in the news**: the number of news pages accessed.
- Measurement of **ideological diversity of news diets**: by assigning domain-level ideological scores from Ad Fontes to the news sites individuals visit and then calculating the ideological distance among those sites; robustness checks using audience-based ideological measures yield consistent results.

d. Analysis

- Linear mixed-effects models (multilevel models)

3. Results and Conclusions

a. Answers to RQs and Hypotheses

- Exposure to unreliable content is embedded in higher overall news consumption.
- Exposure to misinformation is associated with more ideologically diverse news diets.
 1. Misinformation is not crowding out more reliable sources; instead, panelists with larger and more diverse news diets are also more likely to access unreliable content.
- The puzzle is robust to demographic controls and partisan leaning.
 1. Older individuals exhibit higher exposure to unreliable content, measured both by the number of pages visited and the time spent on unreliable sources.

b. Include Key Figures/Tables

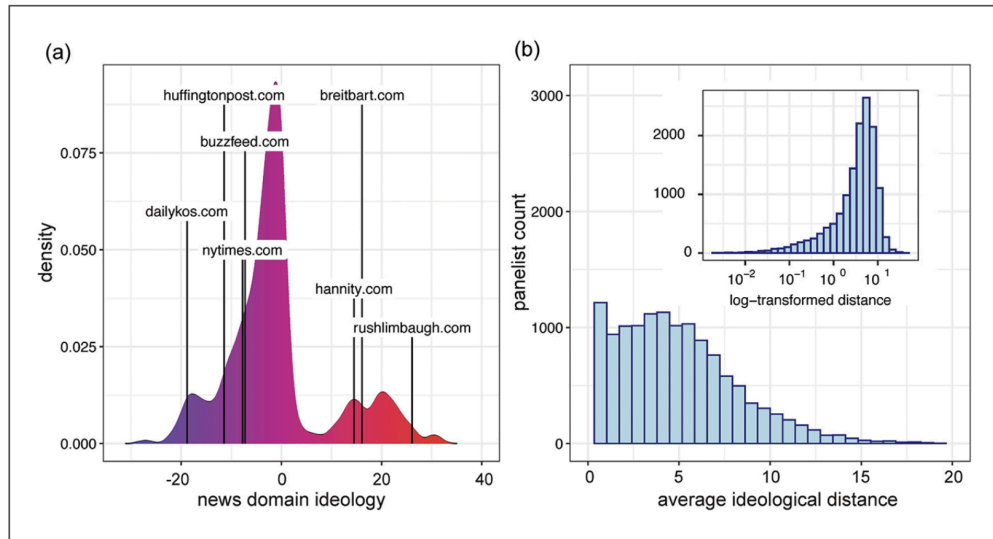


Figure 2. Ideological diversity of news diets. (a) Ideological scores of the news domains visited by the panelists. (b) Ideological diversity of panelists' news diets, measured as the average pairwise distance between the ideology of the news pages they visited (inset shows the log-transformed version of the measure).

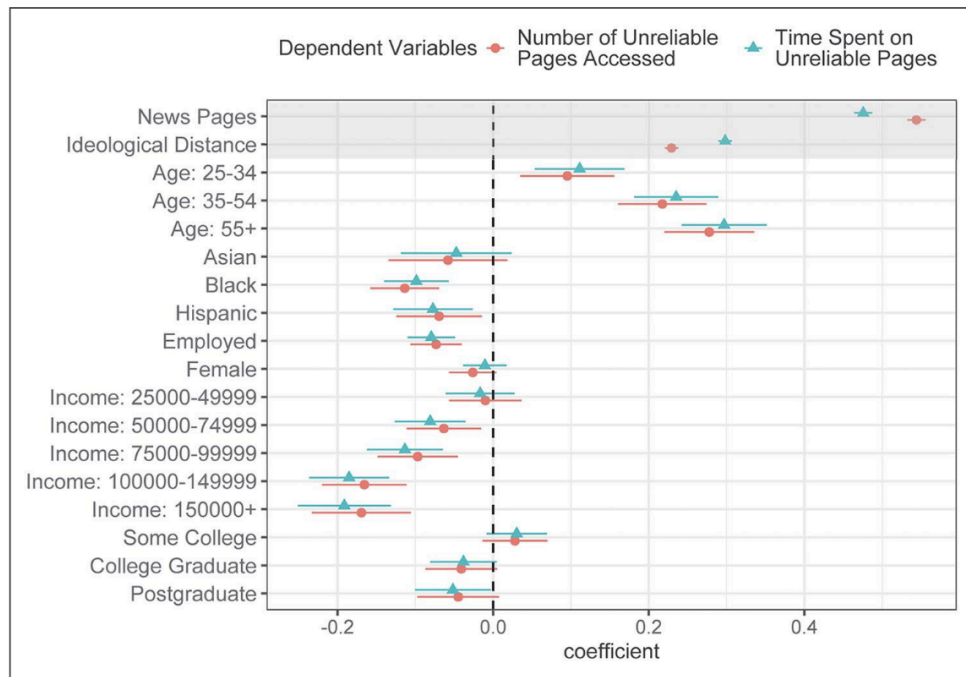


Figure 3. Predictors of exposure to unreliable content.

These estimates result from a linear mixed-effects model with panelist ID and month as random effects. Variables have been scaled (i.e. normalized using the mean value and standard deviation) and log-transformed where appropriate. See Table A1 and A2 in the Supplemental Appendix for additional specifications and details.

c. Authors' Conclusions

- In sum, the study finds that exposure to misinformation is associated with higher overall news consumption and greater ideological diversity,

suggesting that unreliable content is embedded within broader news diets rather than crowding out reliable information.

4. Discussion

a. Implications

- The authors first emphasize that exposure to unreliable content is higher among individuals with greater overall news exposure and more ideologically diverse news diets, rather than among disengaged or ideologically isolated users.
- They suggest that this pattern likely reflects higher levels of political interest, which are associated with more extensive and diverse news consumption. Drawing on prior research, the authors note that political interest is correlated with education and is a key factor underlying political involvement. Taken together, these findings help characterize the population most exposed to misinformation as politically engaged and information-seeking, rather than passive or uninformed.
- This characterization has implications for how scholars and practitioners think about the effects of misinformation and the design of interventions, which should account for the fact that misinformation exposure often occurs among highly engaged news consumers.

b. Limitations

- Generalizability
 1. Prior misinformation research often measures engagement on social media, whereas this study measures exposure through web browsing; because these capture different behaviors, results are not directly comparable across studies.
 2. The data capture desktop-only web browsing, excluding exposure on mobile devices or within apps, which may limit generalizability.
 3. Misinformation exposure is identified through URL visits, which primarily capture text-based web content and do not include misinformation circulating in social media feeds or other digital platforms.
 4. The study does not capture visual or multimodal misinformation (e.g., memes, videos, images), which prior research shows to be prevalent and potentially distinct from text-based misinformation.
- Causality
 1. The study relies on an observational design, which does not allow for causal inference. As a result, it cannot determine whether higher news consumption leads to greater exposure to unreliable content, or whether exposure to unreliable content influences news consumption patterns.
- Measurement at the domain level
 1. Both source reliability and ideological orientation are measured at the news-domain level, not at the level of individual articles or posts. This approach may obscure within-site variation, where reliable and unreliable content may coexist on the same domain.

- Temporal and contextual scope
 1. The data are drawn from one country (the United States) and one time period. Changes in the media environment, platform design, and misinformation since then may limit the applicability of the findings to other contexts or time periods.

c. Future Directions(Inferred from Limitations)

- This study focused on the observational data, and future research could benefit from incorporating exposure through mobile devices.
- The authors suggest that future work could also benefit from designs that bridge exposure-based and engagement-based measures of misinformation.
- The authors highlight the need for future research to account for visual and multimodal misinformation, which is prevalent on many digital platforms but not captured by URL-based measures.
- The study suggests that future research should develop comparable measures of misinformation exposure across platforms..

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Response Letter (~600 words)

- Your assessment of the study
 - **A coherent story**
 - Overall, I find this study particularly strong in how clearly and convincingly it tells a coherent story about misinformation exposure. Rather than focusing on misinformation as isolated content or as a problem to be corrected, the authors take a broader perspective and ask a fundamental question: who is actually encountering misinformation in their everyday news use. In my view, the paper does an excellent job of depicting a concrete user profile of misinformation exposure. The findings suggest that individuals who encounter unreliable content tend to be older, consume more news overall, and draw from more ideologically diverse news sources.
 - **Methodological rigor and scale**
 - Methodologically, the study is especially impressive for its scale and scope. By relying on large-scale, passively collected web-browsing data and tracking users over an entire year, the authors are able to move beyond self-reports or short-term snapshots of behavior. This longitudinal, computational approach allows them to observe stable patterns of exposure across time, which strengthens the credibility of their conclusions. From my perspective, the amount of work required to assemble, clean, and analyze such data is difficult to overstate, particularly for scholars who are more familiar with experiments or small-scale studies. The rigor and labor involved in this type of large-scale computational research are clearly reflected in the quality of the analysis.
 - **Innovative contribution to misinformation research**
 - Finally, I find this study especially valuable because it addresses a dimension of misinformation research that is often under examined. Much of the existing literature focuses on interventions, corrections, or the effectiveness of debunking strategies. This paper instead shifts attention to the descriptive question of who encounters misinformation in the first place. By offering a clear and empirically grounded picture of misinformation exposure profiles, the study provides an important foundation for thinking more carefully about misinformation effects and the design of interventions. In this sense, the paper not only contributes substantially to the literature but also encourages researchers to rethink where misinformation fits within broader patterns of news consumption.
- Weaknesses
 - **Exposure does not imply belief, misperception, or sharing**
 - One additional limitation of this study, beyond those acknowledged by the authors, is that exposure to misinformation does not necessarily translate into belief, misperceptions, or sharing misinformation behavior. Measuring exposure captures whether individuals encounter unreliable content, but it does not indicate how they interpret, evaluate, or respond to

that content. Importantly, individuals differ substantially in how they process misinformation, and these differences are shaped by individual-level factors such as media literacy and news literacy. For example, individuals with higher levels of media or news literacy may be more capable of recognizing low-quality or misleading information and therefore less likely to trust or internalize misinformation, even when they are exposed to it. In this sense, exposure reflects a potential opportunity for influence rather than a direct measure of misinformation's effects. As a result, higher exposure, particularly among individuals with more extensive and diverse news consumption, does not necessarily imply greater harm.

- At the same time, an exposure-based approach may also underrepresent other relevant pathways through which misinformation beliefs form. It is possible that some individuals who hold strong misperceptions or conspiracy beliefs consume relatively little news overall and therefore exhibit low levels of measured exposure, yet nonetheless maintain firm beliefs that are resistant to correction. While this study is not designed to assess such cases, this possibility further underscores the distinction between exposure and belief. As a result, higher exposure, particularly among individuals with more extensive and diverse news consumption, does not necessarily imply greater harm.

o Challenges in measuring misinformation at scale

- A second limitation concerns the broader challenge of measuring misinformation at scale, which this study necessarily confronts. Similar to prior large-scale work in the field (e.g., [Grinberg et al., 2019](#)), the authors identify misinformation exposure at the source level, classifying domains as unreliable based on external credibility assessments. This approach is both common and pragmatic, as content-level identification of misinformation, such as relying on fact-checked claims, is typically feasible only in small-scale or experimental settings.
- However, this measurement strategy also introduces ambiguity. Classifying misinformation at the domain level assumes a degree of homogeneity within sources, even though not all content produced by unreliable outlets is false, and not all content from reliable outlets is accurate. As a result, exposure to “misinformation” in this framework reflects possible contact with lower-credibility information environments rather than verified exposure to false claims. This makes it difficult to assess the substantive consequences of such exposure, particularly in terms of misperceptions or belief formation. More broadly, this limitation highlights a persistent trade-off in misinformation research between measurement precision and scalability, rather than a shortcoming unique to this study.

- Other notes